Multimodal EEG-EMG Integrated Assistive Robotic System

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

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Project Overview

This project focuses on the design and implementation of a hybrid EEG-EMG based robotic arm control system. The system integrates brain signals (EEG) and muscle signals (EMG) to enable intent-driven robotic movement through a complete signal processing and control pipeline.

EEG signals are used to detect high-level user intent through motor imagery, while EMG signals capture muscle activity for execution of movement. These multimodal signals are processed through filtering, feature extraction, and lightweight machine learning algorithms to generate control commands for the robotic arm.

A key feature of the system is the implementation of a dual-signal validation mechanism, referred to as the "Cognitive Gatekeeper." In this approach, robotic actuation is performed only when both EEG-based intent and EMG-based execution signals are simultaneously detected. This improves system reliability by reducing false activations caused by noise in EEG signals and instability in EMG signals.

The system is implemented using low-cost components, including surface bio-signal sensors, microcontroller-based edge computing, and a 3D-printed robotic arm driven by servo motors. Lightweight classifiers such as Linear Discriminant Analysis (LDA) or Support Vector Machines (SVM) are used to ensure real-time performance within hardware constraints.

The developed prototype demonstrates a scalable control system capable of operating a multi-degree-of-freedom robotic arm, validating the effectiveness of hybrid signal fusion for safe and reliable robotic actuation.

Project Statement

Robotic arm control using biological signals remains challenging due to the limitations of single-modality systems. EEG-based control is highly susceptible to noise and false activations, while EMG-based control is affected by muscle fatigue and signal instability.

Therefore, there is a need to develop a reliable and low-cost hybrid EEG-EMG based control system that integrates cognitive intent with muscle activity to achieve accurate and safe robotic arm operation.

Need Analysis

Reliable control of robotic systems using biological signals remains a significant challenge due to limitations of existing unimodal approaches. EEG-based systems, although non-invasive, suffer from low signal resolution, high noise sensitivity, and false activations caused by environmental interference and user artifacts. Conversely, EMG-based systems provide precise and proportional control but are affected by muscle fatigue, electrode displacement, and signal variability over time.

These limitations highlight the need for a hybrid signal processing approach that combines the strengths of both modalities while minimizing their individual weaknesses. By integrating EEG for intent detection and EMG for execution, a more robust and reliable control mechanism can be achieved.

The proposed system addresses this requirement through a dual-signal validation strategy, where robotic actuation is permitted only when both EEG and EMG signals confirm the intended action. This approach improves safety, reduces false triggering, and enhances control stability compared to single-modality systems.

Additionally, implementing such a system using low-cost hardware and edge-based processing enables practical deployment within constrained computational and economic environments. This makes hybrid EEG-EMG control a feasible and effective solution for reliable robotic arm operation.

Literature Survey

The development of a hybrid EEG-EMG based robotic arm control system requires integration of multiple domains, including bio-signal acquisition, signal processing, machine learning, and embedded control. Existing research highlights significant advancements in each of these areas while also exposing key limitations that motivate the proposed work. The effectiveness of any non-invasive brain-computer interface (BCI) system is fundamentally dependent on the quality and reliability of EEG signal acquisition. Early EEG systems were limited to clinical environments, relying on bulky, wired equipment and high-grade amplification systems, which restricted their use in real-time and portable applications. Recent advancements have shifted the focus toward compact, wireless, and edge-based EEG acquisition systems designed for practical deployment. Dou et al. (2026) developed a portable, wireless multi-channel EEG acquisition system optimized for real-time BCI applications. The system integrates analog-to-digital conversion (ADC), signal conditioning, power management, and microcontroller-based processing into a single wearable unit. Experimental evaluation demonstrated high signal quality, with input noise levels below 2.1 μV and a common-mode rejection ratio (CMRR) exceeding 105 dB. The system was successfully validated through a steady-state visual evoked potential (SSVEP) based robotic feeding application, confirming its capability for real-time control. In a related study, Palanichamy et al. (2025) proposed a multimodal approach combining EEG-based motor imagery with real-time gaze tracking to improve robotic control accuracy. Their system achieved a classification accuracy of approximately 75.5%, demonstrating the potential benefits of integrating multiple input modalities for improved reliability. Despite these advancements, several limitations remain. Many modern EEG acquisition systems rely on proprietary architectures, limiting flexibility and customization for research and prototyping. Additionally, multimodal approaches that integrate vision-based systems introduce significant computational overhead and increase system complexity, making them difficult to deploy on low-cost embedded platforms. Therefore, there is a need for lightweight, open-source EEG acquisition systems that provide sufficient signal quality while remaining compatible with real-time processing and resource-constrained hardware environments.

Advanced Deep Learning Architectures for Neural Decoding

Decoding EEG signals is a challenging task due to their non-stationary nature, low amplitude, and susceptibility to noise. Traditional approaches have relied on classical machine learning techniques such as Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM), which offer low computational complexity but limited ability to capture complex temporal patterns in EEG data.Recent advancements have shifted towards deep learning frameworks to improve decoding performance. Ding et al. (2025) proposed the EEG-Deformer, a dense convolutional transformer architecture designed specifically for EEG signal analysis. Unlike conventional Convolutional Neural Networks (CNNs) such as EEGNet, which primarily capture local spatial features, the EEG-Deformer incorporates a Hierarchical Coarse-to-Fine Transformer (HCT) along with a Dense Information Purification (DIP) module. This architecture enables the extraction of multi-scale temporal and frequency-domain features, improving the model's ability to capture long-range dependencies in EEG signals.Experimental evaluations demonstrated that the EEG-Deformer outperformed traditional state-of-the-art models in tasks such as cognitive attention detection and mental workload classification, highlighting the potential of transformer-based approaches in EEG decoding.However, despite their high accuracy, deep learning models such as EEG-Deformer require significant computational resources, including GPU-based processing and large memory capacity. This makes them unsuitable for deployment on low-power embedded systems and microcontrollers used in real-time robotic applications.Therefore, there is a need to adopt lightweight and computationally efficient machine learning techniques that can operate in real-time on edge hardware while maintaining acceptable classification accuracy.

Additive Manufacturing and Low-Cost Myoelectric Control

To address the socio-economic disparities in healthcare access, researchers have increasingly turned to additive manufacturing to bypass the exorbitant costs of traditional prosthetic fabrication. Ali et al. (2026) provided a highly relevant framework detailing the design and development of an affordable myoelectric prosthetic hand tailored for daily functional use. Utilizing Fused Deposition Modeling (FDM) with acrylonitrile butadiene styrene (ABS) plastics, the researchers constructed a

lightweight, human-like structure boasting six degrees of freedom. Actuation was achieved via an artificial tendon-driven mechanism powered by standard DC servo motors, enabling coordinated digit flexion and extension while minimizing mechanical complexity. Early clinical evaluations of the prototype highlighted a critical vulnerability: rudimentary microcontrollers were insufficient for handling real-time signal processing and wireless telemetry concurrently. The system was subsequently upgraded with an Arduino Mega and an ESP8266 Wi-Fi module, drastically improving data transmission and allowing for a complete actuation cycle time of 0.4 seconds with a robust grip force of 3N per finger. The total material cost of the finalized prosthetic was restricted to approximately $250. Despite these mechanical and economic triumphs, the literature consistently underscores that prosthetics reliant strictly on unimodal surface EMG are highly susceptible to signal degradation caused by user perspiration, electrode shift, and unavoidable neuromuscular fatigue, which inevitably compromise control intuitiveness over prolonged use.

Multimodal Signal Fusion and Hybrid Control Architectures

The integration of multiple biological signals has emerged as an effective solution to overcome the limitations of unimodal control systems. Hybrid EEG-EMG architectures combine EEG-based intent detection with EMG-based movement execution, improving reliability and control accuracy.

Bandes Rodriguez and Koike (2025) analyzed different hybrid control strategies, including single-signal, parallel, and sequential approaches. In sequential architectures, EEG is used to determine the intended action, while EMG controls movement execution. Using Linear Discriminant Analysis (LDA), their system achieved classification accuracies above 85%, demonstrating improved performance over unimodal systems.

Various signal fusion techniques have been explored, including feature-level and decision-level fusion. Decision-level fusion, which combines outputs from separate classifiers, has shown better robustness. Rabbani and Islam demonstrated that combining Support Vector Machines (SVM) with temporal models improves classification stability.

Adaptive fusion methods further enhance performance by dynamically adjusting the contribution of EEG and EMG signals based on user conditions such as muscle fatigue. Systems using adaptive Bayesian approaches have achieved accuracies above 90% under fatigue conditions, compared to lower performance in EMG-only systems.

However, many hybrid systems are computationally complex and not suitable for low-cost embedded platforms. Therefore, there is a need for simplified hybrid control architectures that provide reliable performance with low computational overhead.

This motivates the proposed system, which implements a lightweight decision-level fusion approach using a dual-signal validation mechanism ("Cognitive Gatekeeper") to ensure safe and reliable robotic actuation.

Sr. No.	Title	Author	Year	Journal	Main Focus	Technology Used	Research Gap
1	A Wireless EEG Acquisition System Capable of Real-Time BCI Applications	Chenlong Dou, Jinhong Luo, Jidong Lv, Dong Wang, Bochuang Zhao, and Ling Zou	2026	IEEE Sensors Journal	A low-cost, flexible, real-time EEG + BCI platform	EEG, ADC, ESP Microcontroller	Closed architecture,Limited flexibility for research
2	Design and Development of a Low-Cost EMG-Controlled Prosthetic Hand	Fakeraldeen Mohamed Abdalla Ali	2026	CEM Journal by ADBA Scientific	Designing a functional, affordable myoelectric prosthetic arm	EMG, 3D printing, Arduino Mega	High costs and limited accessibility of commercial prostheses
3	EEG-Deformer: A Dense Convolutional Transformer for Brain-Computer Interfaces	Yi Ding	2025	IEEE Journal of Biomedical and Health Informatics	Multi-scale EEG decoding using deep learning	CNN + Transformer, Self-Attention	High complexity, limited real-time use
4	Hybrid EEG-EMG Control Scheme for Multiple Degrees of Freedom Upper-Limb Prostheses	Sorelis Isabel Bandes Rodriguez and Yasuharu Koike	2025	Actuators (MDPI, 2025)	Hybrid control for multi-DOF prosthetic	EEG, EMG hybrid signals with LDA classifier	Limited robustness in real-world hybrid control
5	Multimodal Brain-Computer Interface for Robotic Control: Integration of Real-Time Gaze Tracking and EEG-Based Motor Imagery	Chandresh Palanichamy	2025	Medical & Biological Engineering & Computing (Springer)	Hybrid BCI for robotic arm control	EEG, CSP, LDA, Eye-tracking, Unity	Low accuracy, no real-world testing

Objectives

The primary objective of this project is to develop a hybrid EEG-EMG based robotic arm control system as a proof-of-concept for multimodal signal processing and safe actuation. The specific objectives are:

- **Design a Hybrid Control System:** Develop a dual-modality architecture that integrates EEG-based intent detection with EMG-based movement execution. Implement a dual-signal validation mechanism ("Cognitive Gatekeeper") to ensure actuation only when both signals agree.
- **Implement Signal Processing and Classification:** Apply lightweight signal preprocessing and feature extraction techniques (e.g., filtering, PSD, RMS), along with machine learning models such as LDA or SVM, to classify user intent in real-time on embedded hardware.

- **Develop a Robotic Arm Prototype:** Design and fabricate a low-cost 2-DoF robotic arm using 3D printing and servo motors, capable of performing basic movements such as pick-and-place operations.
- **Ensure Low-Cost and Real-Time Operation:** Implement the complete system on microcontroller-based edge hardware within a constrained budget (~₹15,000), ensuring real-time performance without reliance on external computing.

Methodology

Assumptions

- **Signal Generation:** It is assumed that the user can generate detectable EEG signals through motor imagery and measurable EMG signals through basic muscle contractions.
- **Sensor Placement Stability:** EEG electrodes are assumed to be placed over motor cortex regions (C3, C4, Cz), and EMG sensors over relevant muscles, with stable signal acquisition during operation.
- **Controlled Environment:** System testing and calibration are performed in a controlled indoor environment with minimal electrical noise and motion artifacts.
- **Limited Task Scope:** The robotic arm is designed to perform basic operations such as pick-and-place and simple object manipulation.

Constraints

- **Hardware and Cost Constraints:** The system is implemented using low-cost, open-source components (Arduino, EMG/EEG sensors, 3D-printed parts) within a budget of ₹15,000.
- **Computational Constraints:** Limited processing power of microcontrollers restricts the use of complex deep learning models. Lightweight algorithms such as LDA and SVM are used for real-time execution.

- **Mechanical Limitations:** The robotic arm is constrained by servo motor capabilities, with a maximum payload of approximately 500–600 grams.
- **Control Limitations:** The system operates in an open-loop manner without real-time feedback for force or grip adjustment.

System Architecture and Execution Strategy

The successful engineering of the multimodal robotic arm relies entirely on a highly integrated, sequential processing pipeline. This architecture handles biological data acquisition, rigorous mathematical feature extraction, algorithmic decision-making, and final kinematic actuation, explicitly designed to execute medical-grade control logic using low-cost edge-computing microcontrollers.

1. Non-Invasive Multimodal Signal Acquisition

The operational sequence initiates by simultaneously capturing raw micro-voltage data from two distinct physiological sources, establishing the dual-modality input pipeline completely non-invasively.

- **Cognitive Intent (EEG):** A low-cost 2-Channel Brain BioAmp Band, equipped with highly conductive dry or gel-based electrodes, is precisely positioned over the user's primary motor cortex. This configuration targets the specific cerebral regions responsible for movement planning, capturing the raw electroencephalographic waves that represent the user's focused motor imagery or cognitive intent.
- **Execution Command (EMG):** Surface electromyography sensors, specifically utilizing the BioAmp EXG Pill architecture, are securely affixed to the user's active residual muscle groups (e.g., the flexor carpi ulnaris or biceps brachii). These sensors detect the localized action potentials generated by deliberate muscle depolarizations, representing the user's physical attempt to execute a movement.

2. Hardware Signal Conditioning and Frequency Isolation

Raw biological signals extracted from the skin surface are inherently contaminated by physiological artifacts (such as electrocardiogram interference or ocular blinks) and

environmental electromagnetic noise. Before digital processing can occur, the signals are routed through primary analog amplification and hardware-based filtering stages.

- The EEG signal pathway utilizes precise bandpass filtering to isolate critical brain rhythms, specifically isolating the Mu (8–12 Hz) and Beta (13–30 Hz) frequency bands. These specific bands are highly correlated with the sensorimotor networks, effectively isolating the frequency range where event-related desynchronization occurs during motor imagery.
- The EMG signal pathway employs a broader bandpass filter configuration (typically 20–450 Hz) designed to capture the entire frequency spectrum of rapidly firing muscle fibers while aggressively attenuating low-frequency physical motion artifacts and 50/60Hz mains interference.

3. Real-Time Mathematical Feature Extraction

Once conditioned, the analog voltages are digitized by the Analog-to-Digital Converter (ADC) onboard the primary microcontroller. To reduce data dimensionality and extract meaningful classification patterns, the raw digital streams undergo specific mathematical transformations.

- For the EEG data stream, algorithms compute the Power Spectral Density (PSD), isolating the specific spatial and frequency-domain characteristics of the signal. This process mathematically highlights the event-related desynchronization (ERD) that triggers when a user actively imagines a specific physical movement, providing a quantifiable metric of cognitive intent.
- Simultaneously, for the EMG data stream, the processor continuously calculates the Root Mean Square (RMS) amplitude and waveform length. The RMS calculation provides a smoothed, proportional mathematical envelope that directly correlates to the absolute physical intensity and force of the residual muscle contraction.

4. Edge-Based Machine Learning and The Cognitive Gatekeeper

The mathematically extracted features are subsequently fed into lightweight, pre-trained machine learning classifiers—such as optimized Support Vector Machines (SVM) or Linear Discriminant Analysis (LDA) models—capable of executing rapidly on standard microcontrollers without noticeable latency. This classification stage

categorizes the continuous data into distinct mental states (e.g., 'Intent Confirmed' or 'Idle') and precise physical motor commands (e.g., 'Flexion', 'Extension', 'Pronation').

It is at this critical juncture that the system deploys the **Hybrid Decision-Level Fusion** architecture, governed by the proprietary "Cognitive Gatekeeper" protocol. This logic operates as a strict conditional Boolean AND-gate:

```
IF (EMG_Classifier == Movement_Trigger) AND (EEG_Classifier == Validated_Intent) THEN Execute_Kinematic_Command; ELSE Suppress_Command;
```

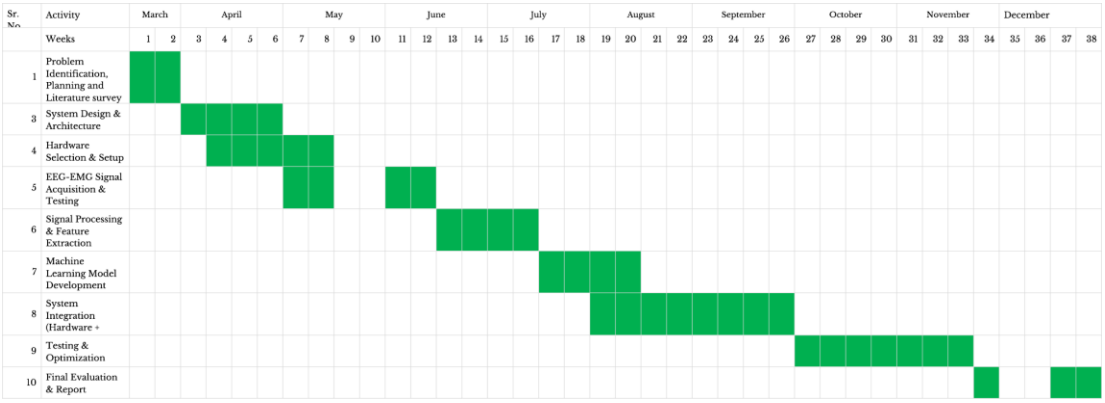
By forcing the system to cross-reference the signals, this mental kill-switch logic drastically reduces the false-positive actuation rate prevalent in highly noisy unimodal systems. The robotic arm is mechanically permitted to move *only* when the user's deliberate physical muscular effort is simultaneously verified by an intense, focused cognitive intent originating from the motor cortex.

5. Inverse Kinematics and Mechatronic Actuation

Upon successfully satisfying the conditions of the Gatekeeper protocol, the confirmed action signal is translated into executable digital commands. An Arduino Mega microcontroller processes these commands through an inverse kinematics matrix, rapidly converting the desired 3D spatial coordinates into precise target angles for each individual joint.

These calculated joint angles are output as high-frequency Pulse Width Modulation (PWM) signals, which are transmitted to the actuation network. The physical architecture relies on an artificial tendon-driven mechanism routed intricately through the 3D-printed ABS plastic chassis segments. High-torque commercial servo motors dynamically manipulate the tension on these internal tendons, directly translating the PWM signals into fluid, coordinated, multi-degree-of-freedom robotic movements. This final stage allows the user to seamlessly execute complex functional tasks—such as grasping, rotating, or precisely adjusting an object—closing the loop from initial brainwave generation to ultimate physical manipulation.

Work Plan



Project Outcomes & Individual Roles

1. **Functional Prototype:**
   A working hybrid EEG-EMG controlled robotic arm capable of performing basic movements such as pick-and-place operations.
2. **Integrated Control Pipeline:**
   A complete real-time system integrating signal acquisition, preprocessing, feature extraction, classification, and actuation on embedded hardware.
3. **Signal Processing and Classification:**
   A system that converts EEG and EMG signals into control commands using lightweight machine learning.
4. **Low-Cost Implementation:**
   A prototype developed using open-source hardware and 3D-printed components within a ₹15,000 budget.

Individual Team Roles

To ensure systematic development and utilize specialized domain expertise, project responsibilities are delegated among the team members as follows:

- **Ishan Singh – Signal Processing Lead:** Focused on signal conditioning and optimization, including ADC interfacing and refining bandpass filtering to improve signal clarity and reduce noise.

- **Anishka Chawla - Signal Processing Lead:** Responsible for signal acquisition and preprocessing, including sensor interfacing and implementing filtering techniques to extract clean EEG and EMG signals.

- **Dharampreet – Machine Learning Specialist:** She focuses on developing and optimizing the machine learning models, including feature extraction and implementing efficient classification algorithms such as LDA and SVM for accurate real-time predictions.

- **Divit Dua – Integration Engineer:** He is responsible for integrating hardware and software components, ensuring smooth communication between the signal processing, machine learning outputs, and the robotic arm control system.

- **Garima Dahuja – Hardware Setup & Configuration Lead:** She manages the configuration and setup of all hardware components, including sensors, microcontrollers, and the robotic arm, ensuring proper connections and reliable system operation.

Course Subjects

The conceptualization, computational modeling, and physical execution of this Capstone Project are directly facilitated by the theoretical knowledge and practical engineering skill sets acquired across various professional core and elective subjects within the B.E. Computer Engineering curriculum.

Course Code	Course Name
UML501	Machine Learning
UCS411	Artificial Intelligence
UCS617	Microprocessor-Based Systems Design
UCS301	Data Structures

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